MARINFOLD / EXPERIMENT 281

Generate hypotheses. Learn to synthesize them.

Iterated offline SFT for a longer training horizon; useful diversity is the hypothesis.

TRAINING DOCUMENT · <contacts-v1.multi>

Sequence context → **Sampled hypotheses** later loss weight 0.1 → **<final-prediction>** switch to synthesis → **Reference contacts** loss weight 1.0

Natural stop: marker loss 1. Forced budget: insert between statements, marker loss 0.

References supervise the answer; they are never in the generation prompt. At inference, generate the answer.

1 FORMAT WARM-UP

Current phase

Teach sections, final marker and end.
Hypothesis loss 1.0; 16 bootstrap drafts.
50% ordinary-contact rehearsal.

2 ITERATED SYNTHESIS

After the format gate

Freeze → sample sequential histories → SFT.
Hypotheses 0.1; reference answer 1.0.
Refresh a large corpus; repeat rounds.

3 REJECTION SFT

Once synthesis works

Sample four whole trajectories per protein.
Select by generated final-answer F1.
Keep history; train on reference answer.

PILOT: 1.5B model · 2,023 train / 25 held-out proteins · batch 32 · 2,000 steps

Next: 200 natural + 200 forced completions. Gate: ≥99% valid in each mode; ≥90% natural with ≥2 hypotheses.*

*Nonempty hypotheses. Larger refreshed SFT and rejection training have not started. Snapshot: 10 Sep 2026.

github.com/Open-Athena/MarinFold/issues/281

```
uv run --no-project --with matplotlib python experiments/exp281_models_iterated_sft_and_rejection_fine_tuning/_scripts/build_idea_slide.py
```